



Be Right™

SAFETY DATA SHEET

Issue Date 10-May-2016

Revision Date 23-Sep-2016

Version 3

Page 1 / 21

1. IDENTIFICATION

Product identifier

Product Name COD, TNTPlus™, Ultra Low Range (1.0-60 MG/L) PK/24

Other means of identification

Product Code(s) TNT820

Safety data sheet number M02451

UN/ID no UN1830

Recommended use of the chemical and restrictions on use

Recommended Use Laboratory reagent. Determination of Chemical Oxygen Demand.

Uses advised against None.

Restrictions on use None.

Details of the supplier of the safety data sheet

Manufacturer Address

Hach Company
P.O.Box 389 Loveland, CO 80539 USA
(970) 669-3050

Emergency telephone number

(303) 623-5716 - 24 Hour Service (515)232-2533 - 8am - 4pm CST

Product Information

Chemical Name Not applicable

Formula Not applicable

CAS No Not applicable

Alternate CAS Number Not applicable

NIOSH (RTECS) Number None reported

2. HAZARDS IDENTIFICATION

Classification

Regulatory Status

This chemical is considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

Corrosive to metals	Category 1
Acute toxicity - Oral	Category 4
Acute toxicity - Dermal	Category 3
Acute toxicity - Inhalation (Dusts/Mists)	Category 4
Skin corrosion/irritation	Category 1
Serious eye damage/eye irritation	Category 1
Specific target organ toxicity (repeated exposure)	Category 2

Hazards not otherwise classified (HNOC)

Product Code(s) TNT820

Issue Date 10-May-2016

Version 3

Product Name COD, TNTPlus™, Ultra Low Range (1.0-60 MG/L) PK/24

Revision Date 23-Sep-2016

Page 2 / 21

Data insufficient for GHS classification but significant enough for mention suggests:

CANCER HAZARD. STRONG INORGANIC ACID MISTS CONTAINING SULFURIC ACID CAN CAUSE CANCER.

Inhalation of low concentrations of sulfuric acid may result in airway irritation such as cough and shortness of breath; high concentrations may result in acute effects such as cough.

Label elements

Signal word - Danger



Hazard statements

H290 - May be corrosive to metals

H301 - Toxic if swallowed

H311 - Toxic in contact with skin

H314 - Causes severe skin burns and eye damage

H332 - Harmful if inhaled

H373 - May cause damage to organs through prolonged or repeated exposure

Precautionary statements

P273 - Avoid release to the environment

P280 - Wear protective gloves/protective clothing/eye protection/face protection

P301 + P312 - IF SWALLOWED: Call a POISON CENTER or doctor/physician if you feel unwell

P301 + P330 + P331 - IF SWALLOWED: rinse mouth. Do NOT induce vomiting

P303 + P361 + P353 - IF ON SKIN (or hair): Remove/Take off immediately all contaminated clothing. Rinse skin with water/shower

P304 + P340 - IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

Other Information

Very toxic to aquatic life with long lasting effects

Very toxic to aquatic life

3. COMPOSITION/INFORMATION ON INGREDIENTS

Substance

Not applicable

Mixture

Percent ranges are used where confidential product information is applicable.

Product Code(s) TNT820

Issue Date 10-May-2016

Version 3

Product Name COD, TNTPlus™, Ultra Low Range (1.0-60 MG/L) PK/24

Revision Date 23-Sep-2016

Page 3 / 21

Chemical Name	CAS No	Percent Range	HMRIC #
Sulfuric acid	7664-93-9	50 - 100%	-
Mercuric sulfate	7783-35-9	1 - 5%	-
Sulfuric acid, disilver(1+) salt	10294-26-5	0.1 - 1%	-
Potassium dichromate	7778-50-9	<0.1%	-

4. FIRST AID MEASURES

Description of first aid measures

General advice	In case of accident or unwellness, seek medical advice immediately (show directions for use or safety data sheet if possible).
Eye contact	IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Call a physician immediately.
Skin contact	IF ON SKIN (or hair): Remove/Take off immediately all contaminated clothing. Rinse skin with water/shower. Call a physician immediately.
Inhalation	IF INHALED: Remove person to fresh air and keep comfortable for breathing. Call a physician immediately.
Ingestion	IF SWALLOWED: Rinse Mouth. Do NOT induce vomiting. Call a physician immediately.
Self-protection of the first aider	Use personal protective equipment as required. Ensure that medical personnel are aware of the material(s) involved and take precautions to protect themselves.

Most important symptoms and effects, both acute and delayed

Symptoms See Section 11: TOXICOLOGICAL INFORMATION.

Indication of any immediate medical attention and special treatment needed

Note to physicians Treat symptomatically.

5. FIRE-FIGHTING MEASURES

Suitable Extinguishing Media

Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

Unsuitable extinguishing media Caution: Use of water spray when fighting fire may be inefficient.

Flammable properties

Substance does not burn.

Specific hazards arising from the chemical

The product causes burns of eyes, skin and mucous membranes. Thermal decomposition can lead to release of irritating and toxic gases and vapors. In the event of fire and/or explosion do not breathe fumes.

Hazardous combustion products This material will not burn.

Protective equipment and precautions for firefighters

Water runoff can cause environmental damage. Dike and collect water used to fight fire.

6. ACCIDENTAL RELEASE MEASURES

U.S. Notice Only persons properly qualified to respond to an emergency involving hazardous substances may respond to a spill according to federal regulations (OSHA 29 CFR

Product Code(s) TNT820

Issue Date 10-May-2016

Version 3

Product Name COD, TNTPlus™, Ultra Low Range (1.0-60 MG/L) PK/24

Revision Date 23-Sep-2016

Page 4 / 21

1910.120(a)(v)) and per your company's emergency response plan and guidelines/procedures. See Section 13, Special Instructions for disposal assistance. Outside of the US, only persons properly qualified according to state or local regulations should respond to a spill involving chemicals.

EC Notice

Only persons properly qualified to respond to an emergency involving hazardous substances should respond to a spill involving chemicals. See Section 13, Special Instructions for disposal assistance.

WHMIS Notice

Only persons properly qualified to respond to an emergency involving hazardous substances should respond to a spill involving chemicals. See Section 13, Special Instructions for disposal assistance.

Personal precautions, protective equipment and emergency procedures

Personal precautions

Evacuate personnel to safe areas. Remove all sources of ignition. Do not touch or walk through spilled material. Ventilate affected area. Use personal protective equipment as required.

For emergency responders

Use personal protection recommended in Section 8.

Environmental precautions

Environmental precautions

Avoid release to the environment. See Section 12 for additional ecological information.

Methods and material for containment and cleaning up

Methods for containment

Prevent further leakage or spillage if safe to do so. Dike far ahead of liquid spill for later disposal.

Methods for cleaning up

Take necessary precautions in observance of pertinent physical hazards. Neutralize spill if necessary. Soak up with inert absorbent material. Take up mechanically, placing in appropriate containers for disposal. Clean contaminated surface thoroughly. Dispose of in accordance with local, state and federal regulations or laws.

Emergency Response Guide Number

137

7. HANDLING AND STORAGE

Precautions for safe handling

Advice on safe handling

Use personal protective equipment as required. Avoid contact with skin, eyes or clothing. Do not breathe dust/fume/gas/mist/vapors/spray.

Conditions for safe storage, including any incompatibilities

Storage Conditions

Keep container tightly closed in a dry and well-ventilated place. Keep out of the reach of children. Keep containers tightly closed in a dry, cool and well-ventilated place. Keep in properly labeled containers. Keep/store only in original container.

Flammability class

Not applicable

Incompatible materials

Caustics. Incompatible with strong acids and bases. Incompatible with oxidizing agents.

8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Control parameters

Exposure Guidelines

.

Product Code(s) TNT820

Issue Date 10-May-2016

Version 3

Product Name COD, TNTPlus™, Ultra Low Range (1.0-60 MG/L) PK/24

Revision Date 23-Sep-2016

Page 5 / 21

Chemical Name	ACGIH TLV	OSHA PEL	NIOSH IDLH
Sulfuric acid 50 - 100%	TWA: 0.2 mg/m ³	TWA: 1 mg/m ³ (vacated) TWA: 1 mg/m ³	IDLH: 15 mg/m ³ TWA: 1 mg/m ³
Mercuric sulfate 1 - 5%	TWA: 0.025 mg/m ³ TWA: 0.1 mg/m ³ S*	(vacated) Ceiling: 0.1 mg/m ³	IDLH: 10 mg/m ³ Hg Ceiling: 0.1 mg/m ³ Hg TWA: 0.05 mg/m ³ except Organo alkyls Hg vapor
Sulfuric acid, disilver(1+) salt 0.1 - 1%	TWA: 0.01 mg/m ³	TWA: 0.01 mg/m ³ (vacated) TWA: 0.01 mg/m ³	IDLH: 10 mg/m ³ Ag TWA: 0.01 mg/m ³ Ag
Potassium dichromate <0.1%	TWA: 0.05 mg/m ³	TWA: 5 µg/m ³ (vacated) Ceiling: 0.1 mg/m ³ Ceiling: 0.1 mg/m ³	IDLH: 15 mg/m ³ Cr(VI) TWA: 0.0002 mg/m ³ Cr

Chemical Name	Alberta OEL	British Columbia OEL	Manitoba OEL	New Brunswick OEL	New Foundland & Labrador OEL
Sulfuric acid 50 - 100%	TWA: 1 mg/m ³ STEL: 3 mg/m ³	TWA: 0.2 mg/m ³	TWA: 0.2 mg/m ³	TWA: 1 mg/m ³ STEL: 3 mg/m ³	TWA: 0.2 mg/m ³
Mercuric sulfate 1 - 5%	TWA: 0.025 mg/m ³ SKN*	TWA: 0.025 mg/m ³ TWA: 0.1 mg/m ³ SKN* R	TWA: 0.025 mg/m ³ TWA: 0.1 mg/m ³ SKN*	TWA: 0.025 mg/m ³ SKN*	TWA: 0.025 mg/m ³ TWA: 0.1 mg/m ³ SKN*
Sulfuric acid, disilver(1+) salt 0.1 - 1%	TWA: 0.01 mg/m ³	TWA: 0.01 mg/m ³ STEL: 0.03 mg/m ³	TWA: 0.01 mg/m ³	TWA: 0.01 mg/m ³	TWA: 0.01 mg/m ³
Potassium dichromate <0.1%	TWA: 0.05 mg/m ³ TWA: 0.5 mg/m ³	TWA: 0.025 mg/m ³ Ceiling: 0.1 mg/m ³	TWA: 0.05 mg/m ³	TWA: 0.05 mg/m ³	TWA: 0.05 mg/m ³

Chemical Name	Northwest Territories OEL	Nova Scotia OEL	Nunavut OEL	Ontario TWA	Prince Edward Island OEL
Sulfuric acid 50 - 100%	TWA: 0.2 mg/m ³ STEL: 0.6 mg/m ³	TWA: 0.2 mg/m ³	TWA: 0.2 mg/m ³ STEL: 0.6 mg/m ³	TWA: 0.2 mg/m ³	TWA: 0.2 mg/m ³
Mercuric sulfate 1 - 5%	TWA: 0.025 mg/m ³ TWA: 0.1 mg/m ³ STEL: 0.075 mg/m ³ STEL: 0.3 mg/m ³ SKN*	TWA: 0.025 mg/m ³ TWA: 0.1 mg/m ³ SKN*	TWA: 0.025 mg/m ³ TWA: 0.1 mg/m ³ STEL: 0.075 mg/m ³ STEL: 0.3 mg/m ³ SKN*	TWA: 0.025 mg/m ³ TWA: 0.1 mg/m ³ SKN*	TWA: 0.025 mg/m ³ TWA: 0.1 mg/m ³
Sulfuric acid, disilver(1+) salt 0.1 - 1%	TWA: 0.01 mg/m ³ STEL: 0.03 mg/m ³	TWA: 0.01 mg/m ³	TWA: 0.01 mg/m ³ STEL: 0.03 mg/m ³	TWA: 0.01 mg/m ³	TWA: 0.01 mg/m ³
Potassium dichromate <0.1%	TWA: 0.05 mg/m ³ STEL: 0.15 mg/m ³	TWA: 0.05 mg/m ³	TWA: 0.05 mg/m ³ STEL: 0.15 mg/m ³	TWA: 0.05 mg/m ³	TWA: 0.05 mg/m ³

Chemical Name	Quebec OEL	Saskatchewan OEL	Yukon OEL
Sulfuric acid 50 - 100%	TWA: 1 mg/m ³ STEL: 3 mg/m ³	TWA: 0.2 mg/m ³ STEL: 0.6 mg/m ³	STEL: 1 mg/m ³ TWA: 1 mg/m ³
Mercuric sulfate 1 - 5%	TWA: 0.025 mg/m ³ SKN*	TWA: 0.025 mg/m ³ TWA: 0.1 mg/m ³ STEL: 0.075 mg/m ³ STEL: 0.3 mg/m ³ SKN*	NDF
Sulfuric acid, disilver(1+) salt 0.1 - 1%	TWA: 0.01 mg/m ³	TWA: 0.01 mg/m ³ STEL: 0.03 mg/m ³	STEL: 0.03 mg/m ³ TWA: 0.01 mg/m ³
Potassium dichromate <0.1%	TWA: 0.05 mg/m ³ SKN+	TWA: 0.05 mg/m ³ TWA: 0.5 mg/m ³ STEL: 0.15 mg/m ³ STEL: 1.5 mg/m ³	STEL: 0.1 mg/m ³ TWA: 0.1 mg/m ³

Other Information

Vacated limits revoked by the Court of Appeals decision in AFL-CIO v. OSHA, 965 F.2d 962 (11th Cir., 1992).

Product Code(s) TNT820

Issue Date 10-May-2016

Version 3

Product Name COD, TNTPlus™, Ultra Low Range (1.0-60 MG/L) PK/24

Revision Date 23-Sep-2016

Page 6 / 21

Legend See section 16 for terms and abbreviations

Appropriate engineering controls

Engineering Controls Showers
Eyewash stations
Ventilation systems

Individual protection measures, such as personal protective equipment

Eye/face protection Wear tight sealing safety goggles and/or face protection shield.

Skin and body protection Wear protective gloves and protective clothing.

Respiratory protection In case of insufficient ventilation, wear suitable respiratory equipment.

General Hygiene Considerations Handle in accordance with good industrial hygiene and safety practice. Do not eat, drink or smoke when using this product. Take off all contaminated clothing and wash it before reuse. Wash hands thoroughly after handling. Regular cleaning of equipment, work area and clothing is recommended.

Environmental exposure controls

Prevent product from entering drains. Do not allow into any sewer, on the ground or into any body of water. Local authorities should be advised if significant spillages cannot be contained.

9. PHYSICAL AND CHEMICAL PROPERTIES

Information on basic physical and chemical properties

Physical state Liquid

Gas Under Pressure Not classified according to GHS criteria

Appearance Turbid solution **Color** light orange

Odor Acidic **Odor threshold** No data available

<u>Property</u>	<u>Values</u>	<u>Remarks • Method</u>
Molecular weight	No data available	
pH	< 0.5	
Melting point/freezing point	~ -7 °C / 19 °F	Estimation based on theoretical calculation
Boiling point / boiling range	~ 102 °C / 216 °F	Estimation based on theoretical calculation
Evaporation rate	0.15 (water = 1)	Estimation based on theoretical calculation
Vapor pressure	1.35 mm Hg / 0.18 kPa at 25 °C / 77 °F	Estimation based on theoretical calculation
Vapor density (air = 1)	0.03 (air = 1)	

Product Code(s) TNT820

Issue Date 10-May-2016

Version 3

Product Name COD, TNTPlus™, Ultra Low Range (1.0-60 MG/L) PK/24

Revision Date 23-Sep-2016

Page 7 / 21

Specific gravity (water = 1 / air = 1) No data available

Partition Coefficient (n-octanol/water) Not applicable

Soil Organic Carbon-Water Partition Coefficient Not applicable

Autoignition temperature No data available

Decomposition temperature No data available

Dynamic viscosity No data available

Kinematic viscosity No data available

Solubility(ies)

Water solubility

<u>Water solubility classification</u>	<u>Water solubility</u>	<u>Water Solubility Temperature</u>
Soluble	> 1000 mg/L	25 °C / 77 °F

Solubility in other solvents

<u>Chemical Name</u>	<u>Solubility classification</u>	<u>Solubility</u>	<u>Solubility Temperature</u>
None reported	No information available	No data available	No information available

Other Information

Metal Corrosivity Classified as corrosive to metal according to GHS criteria

GHS Metal Corrosivity Classification Category 1, H290

Steel Corrosion Rate > 6.25 mm/yr / > 0.25 in/yr

Aluminum Corrosion Rate > 6.25 mm/yr / > 0.25 in/yr

Bulk density Not applicable

Explosive properties Not classified according to GHS criteria.

Explosion data Not Flammable, but reacts with most metals to form flammable hydrogen gas. During a fire, corrosive and toxic gases may be generated by thermal decomposition.

Upper explosion limit No data available

Lower explosion limit No data available

Flammable properties Not classified as flammable according to GHS criteria.

Flammability Limit in Air

Upper flammability limit: No data available

Lower flammability limit: No data available

Flash point No data available

Oxidizing properties Not classified according to GHS criteria.

Product Code(s) TNT820

Issue Date 10-May-2016

Version 3

Product Name COD, TNTPlus™, Ultra Low Range (1.0-60
MG/L) PK/24

Revision Date 23-Sep-2016

Page 8 / 21

Reactivity properties

Not classified as self-reactive, pyrophoric, self-heating or emitting flammable gases in contact with water according to GHS criteria.

10. STABILITY AND REACTIVITY

Reactivity properties

Not classified as self-reactive, pyrophoric, self-heating or emitting flammable gases in contact with water according to GHS criteria

Chemical stability

Stable under recommended storage conditions.

Special dangers of the product

None reported

Possibility of Hazardous Reactions

No information available.

Hazardous polymerization

Hazardous polymerization does not occur.

Conditions to avoid

Exposure to light or contamination by organic materials will affect this product's stability. Exposure to air or moisture over prolonged periods.

Incompatible materials

Caustics. Incompatible with strong acids and bases. Incompatible with oxidizing agents.

Hazardous Decomposition Products

Thermal decomposition can lead to release of irritating and toxic gases and vapors.

Explosive properties

Not classified according to GHS criteria. Not Flammable, but reacts with most metals to form flammable hydrogen gas. During a fire, corrosive and toxic gases may be generated by thermal decomposition.

Upper explosion limit

No data available

Lower explosion limit

No data available

Autoignition temperature

No data available

Sensitivity to Static Discharge

None reported

Sensitivity to Mechanical Impact

None reported

11. TOXICOLOGICAL INFORMATION

Information on Likely Routes of Exposure

Product Information

Toxic in contact with skin. Corrosive to skin. Corrosive to eyes.
Harmful if swallowed. Harmful by inhalation.

Product Code(s) TNT820

Issue Date 10-May-2016

Version 3

Product Name COD, TNTPlus™, Ultra Low Range (1.0-60 MG/L) PK/24

Revision Date 23-Sep-2016

Page 9 / 21

Inhalation	Causes burns. Corrosive by inhalation. Avoid breathing dust/fume/gas/mist/vapors/spray. Harmful by inhalation.
Eye contact	Corrosive to the eyes and may cause severe damage including blindness. Causes burns. Corrosive to eyes.
Skin contact	Toxic in contact with skin. Cause severe skin burns and eye damage. Causes burns.
Ingestion	Ingestion causes burns of the upper digestive and respiratory tracts. Harmful if swallowed. Causes burns.
Aggravated Medical Conditions	Eye disorders. Skin disorders. Respiratory disorders.
Toxicologically synergistic products	None known.
Toxicokinetics, metabolism and distribution	See ingredients information below.

Chemical Name	Toxicokinetics, metabolism and distribution
Sulfuric acid (50 - 100%) CAS#: 7664-93-9	The corrosivity of sulfuric acid makes it difficult to assess its effects on metabolism. Its corrosivity is also the main contributor to acute deaths, therefore it is not classified for acute toxicity.
Mercuric sulfate (1 - 5%) CAS#: 7783-35-9	Central nervous system is the most sensitive target for mercury exposure.

Product Acute Toxicity Data

Oral Exposure Route No data available

Dermal Exposure Route No data available

Inhalation (Dust/Mist) Exposure Route No data available

Inhalation (Vapor) Exposure Route No data available

Inhalation (Gas) Exposure Route No data available

The following values are calculated based on chapter 3.1 of the GHS document

ATEmix (oral)	331.79 mg/kg
ATEmix (dermal)	331.00 mg/kg
ATEmix (inhalation-dust/mist)	3.32 mg/L

Ingredient Acute Toxicity Data

Oral Exposure Route

Chemical Name	Endpoint type	Reported dose	Exposure time	Toxicological effects	Key literature references and sources for data
Sulfuric acid, disilver(1+) salt (0.1 - 1%) CAS#: 10294-26-5	Rat LD ₅₀	> 5000 mg/kg	None reported	None reported	Vendor SDS
Potassium dichromate (<0.1%) CAS#: 7778-50-9	Rat LD ₅₀	25 mg/kg	None reported	None reported	ERMA (New Zealand's Environmental Risk Management Authority)
Chemical Name	Endpoint type	Reported dose	Exposure time	Toxicological effects	Key literature references and sources for data
Sulfuric acid (50 - 100%) CAS#: 7664-93-9	Rat LD ₅₀	2140 mg/kg	None reported	None reported	IUCLID (The International Uniform Chemical Information Database)
Mercuric sulfate (1 - 5%) CAS#: 7783-35-9	Mouse LD ₅₀	25 mg/kg	None reported	None reported	RTECS (Registry of Toxic Effects of Chemical Substances)

Product Code(s) TNT820

Issue Date 10-May-2016

Version 3

Product Name COD, TNTPlus™, Ultra Low Range (1.0-60 MG/L) PK/24

Revision Date 23-Sep-2016

Page 10 / 21

Dermal Exposure Route

Chemical Name	Endpoint type	Reported dose	Exposure time	Toxicological effects	Key literature references and sources for data
Potassium dichromate (<0.1%) CAS#: 7778-50-9	Rat LD ₅₀	1170 mg/kg	None reported	None reported	ERMA (New Zealand's Environmental Risk Management Authority)
Chemical Name	Endpoint type	Reported dose	Exposure time	Toxicological effects	Key literature references and sources for data
Mercuric sulfate (1 - 5%) CAS#: 7783-35-9	Rat LD ₅₀	625 mg/kg	None reported	None reported	RTECS (Registry of Toxic Effects of Chemical Substances)

Inhalation (Dust/Mist) Exposure Route

Chemical Name	Endpoint type	Reported dose	Exposure time	Toxicological effects	Key literature references and sources for data
Potassium dichromate (<0.1%) CAS#: 7778-50-9	Rat LC ₅₀	0.094 mg/L	4 hours	None reported	ERMA (New Zealand's Environmental Risk Management Authority)

Inhalation (Vapor) Exposure Route

Chemical Name	Endpoint type	Reported dose	Exposure time	Toxicological effects	Key literature references and sources for data
Sulfuric acid (50 - 100%) CAS#: 7664-93-9	Rat LC ₅₀	0.510 mg/L	None reported	None reported	LOLI
Chemical Name	Endpoint type	Reported dose	Exposure time	Toxicological effects	Key literature references and sources for data
Sulfuric acid (50 - 100%) CAS#: 7664-93-9	Human TD _{Lo}	0.144 mg/L	4 hours	Lungs, Thorax, or Respiration Dyspnea	RTECS (Registry of Toxic Effects of Chemical Substances)

Inhalation (Gas) Exposure Route

No data available

Product Skin Corrosion/Irritation Data

No data available.

Ingredient Skin Corrosion/Irritation Data

Chemical Name	Test method	Species	Reported dose	Exposure time	Results	Key literature references and sources for data
Sulfuric acid (50 - 100%) CAS#: 7664-93-9	Existing human experience	Human	None reported	None reported	Corrosive to skin	HSDB (Hazardous Substances Data Bank)
Mercuric sulfate (1 - 5%) CAS#: 7783-35-9	Existing human experience	Human	None reported	None reported	Skin irritant	GESTIS (Information System on Hazardous Substances of the German Social Accident Insurance)
Sulfuric acid, disilver(1+) salt (0.1 - 1%) CAS#: 10294-26-5	Standard Draize Test	Rabbit	500 mg	4 hours	Not corrosive or irritating to skin	ECHA (The European Chemicals Agency)

Product Code(s) TNT820

Issue Date 10-May-2016

Version 3

Product Name COD, TNTPlus™, Ultra Low Range (1.0-60 MG/L) PK/24

Revision Date 23-Sep-2016

Page 11 / 21

Product Serious Eye Damage/Eye Irritation Data

No data available.

Ingredient Eye Damage/Eye Irritation Data

Chemical Name	Test method	Species	Reported dose	Exposure time	Results	Key literature references and sources for data
Sulfuric acid (50 - 100%) CAS#: 7664-93-9	Existing human experience	Human	None reported	None reported	Corrosive to eyes	HSDB (Hazardous Substances Data Bank)
Mercuric sulfate (1 - 5%) CAS#: 7783-35-9	Existing human experience	Human	None reported	None reported	Eye irritant	GESTIS (Information System on Hazardous Substances of the German Social Accident Insurance)
Sulfuric acid, disilver(1+) salt (0.1 - 1%) CAS#: 10294-26-5	Standard Draize Test	Rabbit	180 mg	None reported	Corrosive to eyes	ECHA (The European Chemicals Agency)

Sensitization Information

Product Sensitization Data

Skin Sensitization Exposure Route

No data available.

Respiratory Sensitization Exposure Route

No data available.

Ingredient Sensitization Data

Skin Sensitization Exposure Route

No data available.

Respiratory Sensitization Exposure Route

No data available.

Chronic Toxicity Information

Product Repeat Dose Toxicity Data

Oral Exposure Route

No data available.

Dermal Exposure Route

No data available.

Inhalation (Dust/Mist) Exposure Route

No data available.

Inhalation (Vapor) Exposure Route

No data available.

Inhalation (Gas) Exposure Route

No data available.

Ingredient Repeat Dose Toxicity Data

Oral Exposure Route

No data available

Dermal Exposure Route

No data available

Inhalation (Dust/Mist) Exposure Route

No data available

Inhalation (Vapor) Exposure Route

Chemical Name	Endpoint type	Reported dose	Exposure time	Toxicological effects	Key literature references and sources for data
---------------	---------------	---------------	---------------	-----------------------	--

Product Code(s) TNT820

Issue Date 10-May-2016

Version 3

Product Name COD, TNTPlus™, Ultra Low Range (1.0-60 MG/L) PK/24

Revision Date 23-Sep-2016

Page 12 / 21

Sulfuric acid (50 - 100%) CAS#: 7664-93-9	Human TC _{Lo}	.003 mg/L	168 days	Musculoskeletal Changes in teeth and supporting structures	RTECS (Registry of Toxic Effects of Chemical Substances)
---	---------------------------	-----------	----------	---	--

Inhalation (Gas) Exposure Route

No data available

Chemical Name	CAS No	ACGIH	IARC	NTP	OSHA
Sulfuric acid	7664-93-9	A2	Group 1	Known	X
Mercuric sulfate	7783-35-9	-	Group 3	-	-
Sulfuric acid, disilver(1+) salt	10294-26-5	-	-	-	-
Potassium dichromate	7778-50-9	A1	Group 1	Known	X

Legend

ACGIH (American Conference of Governmental Industrial Hygienists)	A2 - Suspected Human Carcinogen
IARC (International Agency for Research on Cancer)	Group 1 - Carcinogenic to Humans Not classifiable as a human carcinogen
NTP (National Toxicology Program)	Known - Known Carcinogen
OSHA (Occupational Safety and Health Administration of the US Department of Labor)	X - Present

Product Carcinogenicity Data

No data available

Oral Exposure Route

No data available

Dermal Exposure Route

No data available

Inhalation (Dust/Mist) Exposure Route

No data available

Inhalation (Vapor) Exposure Route

No data available

Inhalation (Gas) Exposure Route

No data available

Ingredient Carcinogenicity Data

Oral Exposure Route

No data available

Dermal Exposure Route

No data available

Inhalation (Dust/Mist) Exposure Route

No data available

Inhalation (Vapor) Exposure Route

No data available

Inhalation (Gas) Exposure Route

No data available

Product Germ Cell Mutagenicity *invitro* Data

No data available.

Ingredient Germ Cell Mutagenicity *invitro* Data

Chemical Name	Test	Cell Strain	Reported dose	Exposure time	Results	Key literature references and sources for data
Sulfuric acid (50 - 100%)	Cytogenetic analysis	Hamster ovary	4 mmol/L	None reported	Positive test result for mutagenicity	No information available

Product Code(s) TNT820

Issue Date 10-May-2016

Version 3

Product Name COD, TNTPlus™, Ultra Low Range (1.0-60 MG/L) PK/24

Revision Date 23-Sep-2016

Page 13 / 21

CAS#: 7664-93-9						
Potassium dichromate (<0.1%) CAS#: 7778-50-9	Micronucleus test	Human lymphocyte	0.3 mg/L	None reported	Positive test result for mutagenicity	RTECS (Registry of Toxic Effects of Chemical Substances)
Chemical Name	Test	Cell Strain	Reported dose	Exposure time	Results	Key literature references and sources for data
Potassium dichromate (<0.1%) CAS#: 7778-50-9	Morphological transformation	Human fibroblast	200 nmol/L	None reported	Positive test result for mutagenicity	RTECS (Registry of Toxic Effects of Chemical Substances)
Chemical Name	Test	Cell Strain	Reported dose	Exposure time	Results	Key literature references and sources for data
Potassium dichromate (<0.1%) CAS#: 7778-50-9	DNA damage	Human fibroblast	500 nmol/L	None reported	Positive test result for mutagenicity	RTECS (Registry of Toxic Effects of Chemical Substances)

Oral Exposure Route

No data available

Dermal Exposure Route

No data available

Inhalation (Dust/Mist) Exposure Route

No data available

Inhalation (Vapor) Exposure Route

No data available

Inhalation (Gas) Exposure Route

No data available

Ingredient Germ Cell Mutagenicity *in vivo* Data

Oral Exposure Route

No data available

Dermal Exposure Route

No data available

Inhalation (Dust/Mist) Exposure Route

No data available

Inhalation (Vapor) Exposure Route

No data available

Inhalation (Gas) Exposure Route

No data available

Oral Exposure Route

No data available

Dermal Exposure Route

No data available

Inhalation (Dust/Mist) Exposure Route

No data available

Inhalation (Vapor) Exposure Route

No data available

Inhalation (Gas) Exposure Route

No data available

Ingredient Reproductive Toxicity Data

Oral Exposure Route

Chemical Name	Endpoint type	Reported dose	Exposure time	Toxicological effects	Key literature references and sources for data
Potassium dichromate (<0.1%) CAS#: 7778-50-9	Mouse TD _{Lo}	1710 mg/kg	19 days	Effects on Embryo or Fetus Fetotoxicity (except death e.g. stunted fetus) Effects on Fertility	RTECS (Registry of Toxic Effects of Chemical Substances)

Product Code(s) TNT820

Issue Date 10-May-2016

Version 3

Product Name COD, TNTPlus™, Ultra Low Range (1.0-60 MG/L) PK/24

Revision Date 23-Sep-2016

Page 14 / 21

				Post-implantation mortality (e.g. dead and/or resorbed implants per total number of implants) Specific Developmental Abnormalities Craniofacial (including nose and tongue)	
Chemical Name	Endpoint type	Reported dose	Exposure time	Toxicological effects	Key literature references and sources for data
Potassium dichromate (<0.1%) CAS#: 7778-50-9	Mouse TD _{Lo}	3420 mg/kg	None reported	Effects on Fertility Litter size (e.g. # fetuses per litter; measured before birth) Specific Developmental Abnormalities Skin and appendages	RTECS (Registry of Toxic Effects of Chemical Substances)

Dermal Exposure Route

No data available

Inhalation (Dust/Mist) Exposure Route

No data available

Inhalation (Vapor) Exposure Route

Chemical Name	Endpoint type	Reported dose	Exposure time	Toxicological effects	Key literature references and sources for data
Sulfuric acid (50 - 100%) CAS#: 7664-93-9	Rabbit TC _{Lo}	.02 mg/L	7 hours	Specific Developmental Abnormalities Musculoskeletal system	No information available

Inhalation (Gas) Exposure Route

No data available

12. ECOLOGICAL INFORMATION

Ecotoxicity

Very toxic to aquatic life with long lasting effects.

Product Ecological Data

Aquatic toxicity

Fish

No data available

Crustacea

No data available

Algae

No data available

Terrestrial toxicity

Soil

No data available

Vertebrates

No data available

Invertebrates

No data available

Ingredient Ecological Data

Aquatic toxicity

Fish

Chemical Name	Exposure time	Species	Endpoint type	Reported dose	Key literature references and sources for data
Sulfuric acid, disilver(1+) salt	96 hours	<i>Pimephales promelas</i>	LC ₅₀	0.0012 mg/L	GESTIS (Information System on Hazardous Substances of the

Product Code(s) TNT820

Issue Date 10-May-2016

Version 3

Product Name COD, TNTPlus™, Ultra Low Range (1.0-60 MG/L) PK/24

Revision Date 23-Sep-2016

Page 15 / 21

(0.1 - 1%) CAS#: 10294-26-5					German Social Accident Insurance)
Potassium dichromate (<0.1%) CAS#: 7778-50-9	96 hours	<i>Lepomis macrochirus</i>	LC ₅₀	0.131 mg/L	ERMA (New Zealand's Environmental Risk Management Authority)
Chemical Name	Exposure time	Species	Endpoint type	Reported dose	Key literature references and sources for data
Sulfuric acid (50 - 100%) CAS#: 7664-93-9	96 hours	<i>Lepomis macrochirus</i>	LC ₅₀	> 16 mg/L	IUCLID (The International Uniform Chemical Information Database)
Mercuric sulfate (1 - 5%) CAS#: 7783-35-9	7 days	<i>Oncorhynchus gorbuscha</i>	LC ₅₀	0.14 mg/L	EPA (United States Environmental Protection Agency)
Chemical Name	Exposure time	Species	Endpoint type	Reported dose	Key literature references and sources for data
Sulfuric acid, disilver(1+) salt (0.1 - 1%) CAS#: 10294-26-5	217 days	<i>Salmo trutta</i>	EC ₁₀	0.00019 mg/L	GESTIS (Information System on Hazardous Substances of the German Social Accident Insurance)

Crustacea

Chemical Name	Exposure time	Species	Endpoint type	Reported dose	Key literature references and sources for data
Sulfuric acid, disilver(1+) salt (0.1 - 1%) CAS#: 10294-26-5	48 Hours	<i>Daphnia magna</i>	LC ₅₀	0.00022 mg/L	GESTIS (Information System on Hazardous Substances of the German Social Accident Insurance)
Potassium dichromate (<0.1%) CAS#: 7778-50-9	48 Hours	<i>Daphnia magna</i>	EC ₅₀	0.035 mg/L	ERMA (New Zealand's Environmental Risk Management Authority)
Chemical Name	Exposure time	Species	Endpoint type	Reported dose	Key literature references and sources for data
Sulfuric acid (50 - 100%) CAS#: 7664-93-9	48 hours	<i>Crangon crangon</i>	EC ₅₀	> 70 mg/L	IUCLID (The International Uniform Chemical Information Database)
Sulfuric acid, disilver(1+) salt (0.1 - 1%) CAS#: 10294-26-5	48 hours	<i>Ceriodaphnia dubia</i>	EC ₅₀	0.0045 mg/L	GESTIS (Information System on Hazardous Substances of the German Social Accident Insurance)

Algae

Chemical Name	Exposure time	Species	Endpoint type	Reported dose	Key literature references and sources for data
Mercuric sulfate (1 - 5%) CAS#: 7783-35-9	14 days	<i>Pseudokirchnerella subcapitata</i>	EC ₅₀	0.033 mg/L	EPA (United States Environmental Protection Agency)

Terrestrial toxicity

Soil

No data available

Vertebrates

No data available

Invertebrates

No data available

Other Information

Product Code(s) TNT820

Issue Date 10-May-2016

Version 3

Product Name COD, TNTPlus™, Ultra Low Range (1.0-60 MG/L) PK/24

Revision Date 23-Sep-2016

Page 16 / 21

**Canadian Environmental Protection Act (CEPA) - Domestic Substances List (DSL):
Environmentally Hazardous Substances Categorizations**

Chemical Name	CAS No	Category	Persistent	Bioaccumulation	Inherently Toxic to Aquatic Organisms
Sulfuric acid	7664-93-9	-	-	-	-
Mercuric sulfate	7783-35-9	-	-	-	-
Sulfuric acid, disilver(1+) salt	10294-26-5	-	-	-	-
Potassium dichromate	7778-50-9	-	-	-	-

Persistence and degradability

None known.

Product Biodegradability Data

If available, see ingredient data below.

Ingredient Biodegradability Data

Test data reported below

Chemical Name	Test method	Biodegradation	Exposure time	Results
Mercuric sulfate (1 - 5%) CAS#: 7783-35-9	None reported	None reported	None reported	Not readily biodegradable
Sulfuric acid, disilver(1+) salt (0.1 - 1%) CAS#: 10294-26-5	Inorganic Salt	None reported	None reported	Not readily biodegradable

Bioaccumulation

If available, see ingredient data below.

Product Bioaccumulation Data

If available, see ingredient data below.

Ingredient Bioaccumulation Data

Chemical Name	Test method	Exposure time	Species	Bioconcentration factor (BCF)	Results
Mercuric sulfate (1 - 5%) CAS#: 7783-35-9	None reported	None reported	None reported	BCF > 1000	Has the potential to bioaccumulate
Sulfuric acid, disilver(1+) salt (0.1 - 1%) CAS#: 10294-26-5	None reported	8 days	<i>Oncorhynchus mykiss</i>	BCF = 2.5	Does not have the potential to bioaccumulate

Additional information

Product Information

Partition Coefficient (n-octanol/water)

Not applicable

Ingredient Information

Product Code(s) TNT820

Issue Date 10-May-2016

Version 3

Product Name COD, TNTPlus™, Ultra Low Range (1.0-60 MG/L) PK/24

Revision Date 23-Sep-2016

Page 17 / 21

Chemical Name	Partition Coefficient (n-octanol/water)	Method
Sulfuric acid, disilver(1+) salt (0.1 - 1%) CAS#: 10294-26-5	log K _{ow} > 6.18	Estimation through KOWWIN v1.68 part of the Estimation Programs Interface (EPI) Suite™

Mobility

Mobility in soil: High mobility. If available, see ingredient data below.

Product Information

Soil Organic Carbon-Water Partition Coefficient Not applicable

Ingredient Information

Chemical Name	Soil Organic Carbon-Water Partition Coefficient	Method
Sulfuric acid, disilver(1+) salt (0.1 - 1%) CAS#: 10294-26-5	log K _{oc} > 4.83	No information available

Additional information

Water solubility

Product Information

<u>Water solubility classification</u>	<u>Water solubility</u>	<u>Water Solubility Temperature</u>
Soluble	> 1000 mg/L	25 °C / 77 °F

Ingredient Information

Chemical Name	Water solubility classification	Water solubility	Water solubility temperature °C	Water solubility temperature °F
Sulfuric acid CAS#: 7664-93-9	Soluble	> 1000 mg/L	25 °C	77 °F
Sulfuric acid, disilver(1+) salt CAS#: 10294-26-5	Soluble	8000 mg/L	20 °C	68 °F
Potassium dichromate CAS#: 7778-50-9	Moderately soluble	430 mg/L	0 °C	32 °F

Other adverse effects

Contains a substance with an endocrine-disrupting potential.

13. DISPOSAL CONSIDERATIONS

Waste treatment methods

Disposal of wastes Disposal should be in accordance with applicable regional, national, and local laws and regulations.

Contaminated packaging Do not reuse container.

US EPA Waste Number D002

Product Code(s) TNT820

Issue Date 10-May-2016

Version 3

Product Name COD, TNTPlus™, Ultra Low Range (1.0-60 MG/L) PK/24

Revision Date 23-Sep-2016

Page 18 / 21

Special instructions for disposal Dispose of all mercury contaminated material at an E.P.A. hazardous waste facility.
Dispose of material in an E.P.A. approved hazardous waste facility.

14. TRANSPORT INFORMATION

DOT

UN/ID no	UN1830
Proper shipping name	Sulphuric Acid
Hazard Class	8
Packing Group	II
Marine pollutant	This product contains a chemical which is listed as a severe marine pollutant according to DOT.
Emergency Response Guide Number	137

TDG

UN/ID no	UN1830
Proper shipping name	Sulphuric Acid
Hazard Class	8
Packing Group	II
Marine pollutant	This product contains a chemical which is listed as a severe marine pollutant according to TDG. Lead compounds.

IATA

UN/ID no	UN1830
Proper shipping name	Sulphuric Acid
Hazard Class	8
Packing Group	II
ERG Code	137

IMDG

UN/ID no	UN1830
Proper shipping name	Sulphuric Acid
Hazard Class	8
Packing Group	II
Marine pollutant	This material meets the definition of a marine pollutant

Additional information

There is a possibility that this product could be contained in a reagent set or kit composed of various compatible dangerous goods.

If the item is not in a reagent set or kit, the classification given above applies.

If the item is part of a reagent set or kit the classification would change to the following:

UN3316 Chemical Kit, Hazard Class 9, Packing Group II or III.

If the item is not regulated, the Chemical Kit classification does not apply.

15. REGULATORY INFORMATION

National Inventories

TSCA	Complies
DSL/NDSL	Complies

TSCA- United States Toxic Substances Control Act Section 8(b) Inventory

DSL/NDSL- Canadian Domestic Substances List/Non-Domestic Substances List

International Inventories

EINECS/ELINCS	Complies
ENCS	Complies
IECSC	Complies
KECL	Complies
PICCS	Complies

Product Code(s) TNT820

Issue Date 10-May-2016

Version 3

Product Name COD, TNTPlus™, Ultra Low Range (1.0-60 MG/L) PK/24

Revision Date 23-Sep-2016

Page 19 / 21

TCSI Complies
AICS Complies
NZIoC Complies

EINECS/ELINCS- European Inventory of Existing Chemical Substances/European List of Notified Chemical Substances

ENCS- Japan Existing and New Chemical Substances

IECSC- China Inventory of Existing Chemical Substances

KECL- Korean Existing and Evaluated Chemical Substances

PICCS- Philippines Inventory of Chemicals and Chemical Substances

TCSI- Taiwan Chemical Substances Inventory

AICS- Australian Inventory of Chemical Substances

NZIoC- New Zealand Inventory of Chemicals

US Federal Regulations

SARA 313

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product contains a chemical or chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372

Chemical Name	SARA 313 - Threshold Values %
Sulfuric acid (CAS #: 7664-93-9)	1.0
Mercuric sulfate (CAS #: 7783-35-9)	1.0
Sulfuric acid, disilver(1+) salt (CAS #: 10294-26-5)	1.0
Potassium dichromate (CAS #: 7778-50-9)	0.1

SARA 311/312 Hazard Categories

Acute health hazard	Yes
Chronic Health Hazard	Yes
Fire hazard	No
Sudden release of pressure hazard	No
Reactive Hazard	No

CWA (Clean Water Act)

This product does not contain any substances regulated as pollutants pursuant to the Clean Water Act (40 CFR 122.21 and 40 CFR 122.42)

Chemical Name	CWA - Reportable Quantities	CWA - Toxic Pollutants	CWA - Priority Pollutants	CWA - Hazardous Substances
Sulfuric acid 7664-93-9	1000 lb	-	-	X
Mercuric sulfate 7783-35-9	10 lb	X	-	X
Sulfuric acid, disilver(1+) salt 10294-26-5	-	X	-	-
Potassium dichromate 7778-50-9	10 lb	X	-	X

CERCLA

This material, as supplied, does not contain any substances regulated as hazardous substances under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302) or the Superfund Amendments and Reauthorization Act (SARA) (40 CFR 355). There may be specific reporting requirements at the local, regional, or state level pertaining to releases of this material

Chemical Name	Hazardous Substances RQs	CERCLA/SARA RQ	Reportable Quantity (RQ)
Sulfuric acid 7664-93-9	1000 lb	1000 lb	RQ 1000 lb final RQ RQ 454 kg final RQ
Mercuric sulfate 7783-35-9	10 lb	-	RQ 10 lb final RQ RQ 4.54 kg final RQ

Product Code(s) TNT820

Issue Date 10-May-2016

Version 3

Product Name COD, TNTPlus™, Ultra Low Range (1.0-60 MG/L) PK/24

Revision Date 23-Sep-2016

Page 20 / 21

Potassium dichromate 7778-50-9	10 lb	-	RQ 10 lb final RQ RQ 4.54 kg final RQ
-----------------------------------	-------	---	--

U.S. - DEA (Drug Enforcement Administration) List I & List II

Chemical Name	U.S. - DEA (Drug Enforcement Administration) - List I or Precursor Chemicals	U.S. - DEA (Drug Enforcement Administration) - List II or Essential Chemicals
Sulfuric acid (50 - 100%) CAS#: 7664-93-9	Not Listed	50 gallon Export Volume (Exports, transshipments and international transactions to designated countries)

US State Regulations

California Proposition 65

This product contains the following Proposition 65 chemicals

Chemical Name	California Proposition 65
Mercuric sulfate (CAS #: 7783-35-9)	Developmental
Potassium dichromate (CAS #: 7778-50-9)	Carcinogen Developmental Female Reproductive Male Reproductive

IMERC: Contains Mercury Dispose of in accordance with local, state and federal regulations or laws.

U.S. State Right-to-Know Regulations

Chemical Name	New Jersey	Massachusetts	Pennsylvania
Sulfuric acid 7664-93-9	X	X	X
Mercuric sulfate 7783-35-9	X	X	X
Sulfuric acid, disilver(1+) salt 10294-26-5	X	-	X
Potassium dichromate 7778-50-9	X	X	X

U.S. EPA Label Information

EPA Pesticide Registration Number Not applicable

Canada - CEPA - Mercury Containing Products

Chemical Name	Canada - CEPA - Mercury Containing Products
Mercuric sulfate CAS#: 7783-35-9	Applies

16. OTHER INFORMATION, INCLUDING DATE OF PREPARATION OF THE LAST REVISION

Special Comments

This product contains mercury and may be subject to reporting and recordkeeping requirements

NFPA and HMIS Classifications

NFPA	Health hazards - 3	Flammability - 0	Instability - 0	Physical and Chemical Properties -
HMIS	Health hazards - 3	Flammability - 0	Physical hazards - 0	Personal protection - X

Product Code(s) TNT820

Issue Date 10-May-2016

Version 3

Product Name COD, TNTPlus™, Ultra Low Range (1.0-60
MG/L) PK/24

Revision Date 23-Sep-2016

Page 21 / 21

				- See section 8 for more information
--	--	--	--	--------------------------------------

Key or legend to abbreviations and acronyms used in the safety data sheet

NIOSH IDLH

Immediately Dangerous to Life or Health

ACGIH

ACGIH (American Conference of Governmental Industrial Hygienists)

NDF

no data

Legend - Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

TWA	TWA (time-weighted average)	STEL	STEL (Short Term Exposure Limit)
MAC	Maximum Allowable Concentration	Ceiling	Ceiling Limit Value
X	Listed	Vacated	These values have no official status. The only binding levels of contaminants are those listed in the final OSHA PEL. These lists are for reference purposes only. Please note that some reference state regulations of these "liberated" exposure limits in their state regulations.
SKN*	Skin designation	SKN+	Skin sensitization
RSP+	Respiratory sensitization	**	Hazard Designation
C	Carcinogen	R	Reproductive toxicant
M	mutagen		

Prepared By Hach Product Compliance Department

Issue Date 10-May-2016

Revision Date 23-Sep-2016

Revision Note None

Disclaimer

USER RESPONSIBILITY: Each user should read and understand this information and incorporate it in individual site safety programs in accordance with applicable hazard communication standards and regulations.

THE INFORMATION CONTAINED HEREIN IS BASED ON DATA CONSIDERED TO BE ACCURATE. HOWEVER, NO WARRANTY IS EXPRESSED OR IMPLIED REGARDING THE ACCURACY OF THESE DATA OR THE RESULTS TO BE OBTAINED FROM THE USE THEREOF.

HACH COMPANY©2016

End of Safety Data Sheet